AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

CSC4118: COMPUTER GRAPHICS

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Section: [O]

Group No: 03

Project Report On

Project Name [War Zone]

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Table of Contents

SL No	Descriptions	page
1	Title Page	01
2	Table of Contents	02
3	Introduction	03
4	Project Graph	03
5	List of Objects with Object IDs	04
6	List of Functions to Represent Objects	04
7	List of Animation Functions with IDs	05
8	Contribution	05
9	Conclusion	05

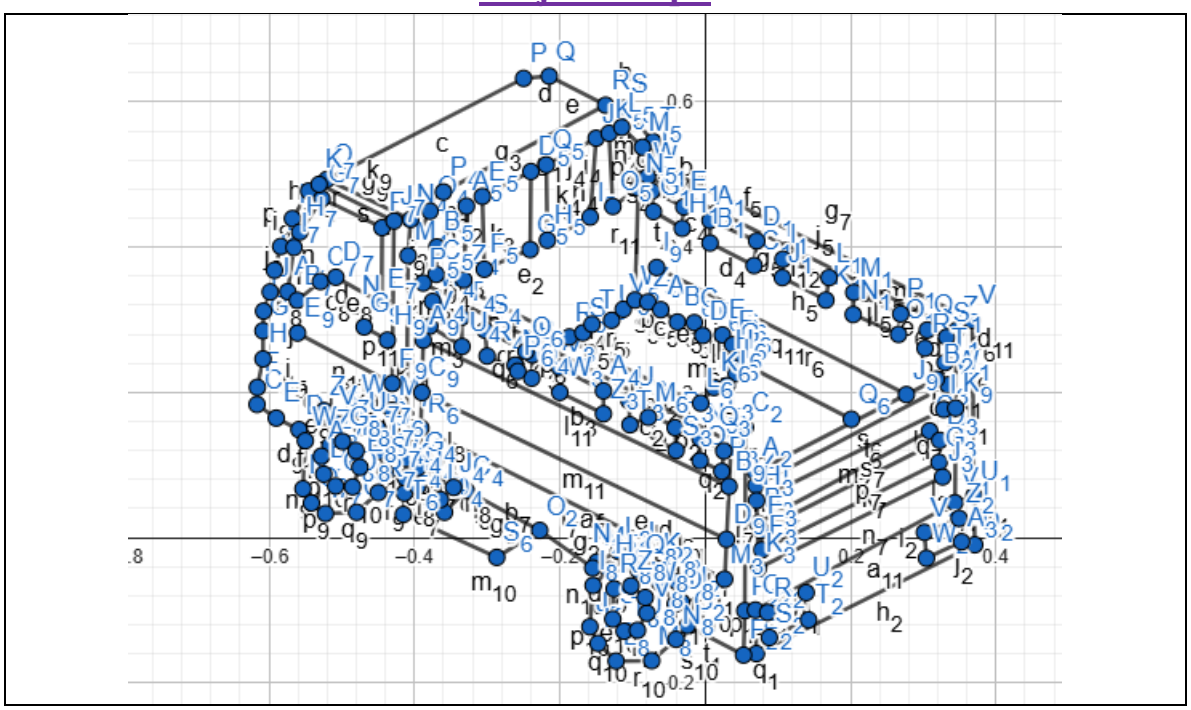
Introduction

The city is no longer peaceful. Smoke rises from shattered buildings. Fires burn quietly among the ruins. A lone tank rolls down the street, moving cautiously towards the city. It stops. Its cannon fires — a bright flash, a loud boom, and the ground shakes. Above, a military helicopter is hovering. Its rotor blades spin loudly as it scans the streets below. In the distance, a missile streaks through the sky, launched from somewhere hidden. Moments later, it strikes. The explosion lights up the skyline. A building collapses. Amidst the chaos, civilians flee. A truck speeds through the rain-soaked roads — one of the last vehicles trying to escape the destruction. The rain pours. It doesn't stop the war, but it hides the tears of the city. A blinking road light gives them hope.

Technologies used:

≫ C++
≫ OpenGL (GL, GLUT)
≫ Windows Platform (Win32)

Project Graph



The project works on a 2D coordinate system with approximate coordinate ranges:

≫ X-axis: -1 to 1
≫ Y-axis: -1 to 1

Dynamic objects like cars, buses, trains, planes, and clouds move within this coordinate framework and Textures has been added.

List of Objects Assigning an Object ID

SL#	Object ID	Object Name
1	OBJ1	Tank
2	OBJ2	Tank Fire
3	OBJ3	Missile
4	OBJ4	Copper
5	OBJ5	Artillery gun
6	OBJ6	Artillery gun Fire
7	OBJ7	Truck
8	OBJ8	Blast
9	OBJ9	Rain
10	OBJ10	Thunder
11	OBJ11	Light Cone Ray
12	OBJ12	Jeep/New Tank

List of Functions To Represent Objects:

SL#	Object Name	Function Name
1	Rain	drawRain()
2	Light Cone	drawLightCone()
3	Background, Gun Fire, Artillery gun, Missile, Copper and Rotor	display()
4	Jeep	drawJeep()

List of Animation Functions with ID

SL#	Animation Function ID	Animation Function	Object/Scene
1	ANM1	updateRain()	Rain
2	ANM2	toggleLightCone()	Light Cone
3	ANM3	triggerCameraShake()	Camera Shaking
4	ANM4	startExplosion()	Explosions
5	ANM5	update ()	Truck, Switch Scenes, Gun Fire, Missile, Copper and Rotor, jeep.
6	ANM6	updateThunder()	Thunder

Contribution

Member Name	Implemented Functions	Implemented Animation Functions	Percentage of Contribution
Navin, Md. Nawshin	All drawing functions, and objects	All animation functions	100%

Conclusion

This project successfully integrates various static and dynamic objects to simulate a lively war zone environment using OpenGL. Features like day-rainy night transitions, gun fires, and multiple moving vehicles provide realism and an interactive feeling. The project also demonstrates strong modular design principles, using separate functions for each object and animation.